

DIODE SEMICONDUCTOR MODULES

35 years

Designed in Germany - Made with Pride in India

RECTIFIER BRIDGES	Current	Voltage	Part Number	
	25 A	200 V – 1000 V	PSB 25MB & PSD 25MT PSB 25MBN & PSD 25MTN	
		1200 V – 2200 V	PSB 25T & PSD 25T PSB 25TN & PSD 25TN	
	36 A	200 V – 1000 V	PSB 36MB & PSD 36MT PSB 36MBN & PSD 36MTN	
		1200 V – 2200 V	PSB 36T & PSD 36T PSB 36TN & PSD 36TN	

Highlights:

- > Single phase and three phase uncontrolled bridges
- > Compact package, easy to mount with one screw
- > Connectors suitable for easy PCB mounting
- > Gold plated or nickel plated terminals
- > UL certified, RoHS and REACH conform and Halogen-Free

Nomenclature:

TN	Add On T, MT, MB = Different chip categories, Gold Plated TN, MTN, MBN = Different chip categories, Nickel Plated
25	Current Rating 25 = 25A 36 = 36A
PSD	Device Type PSD – Three Phase PSB – Single Phase

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body hacks into the device, it should be tough for the person to extract the data. But that might result in more cost of the device and more processing power requirement.

“You have to look at the ROI (return on investment) of the application that you are thinking of. One should not apply IoT just anywhere. It should make sense from a use-case perspective. Same goes for AI on edge. If you want to make the edge smarter, then consider whether it is an ROI-friendly decision, because if you need to invest to make every device smarter, then with that there will be added complexity of making the solution secure. One needs to be clear whether any extra investment is going to justify the output or the ROI that will be derived from the solution. It all depends upon what kind of real-time decision you need to make at the edge, or whether sending the feed to the cloud or network is going to be costly, time-consuming, or not secure,” advises Vinay Solanki.

Amit Kuhrekar advises having

sufficient whenever you are developing something. Consider the cost of development, cost of hardware, and how many such devices are going to be needed. At the end of the day, you will have to maintain that.

“Always factor in the cost of your solution in the design process itself,” opines Nikhil Bhaskaran, founder, Shunya OS.

In terms of cost-effectiveness, it is a matter of business decision for having an aggregated data model in a manufacturing unit. If you have twenty machines, then it is better to deploy sensors at each machine, which may be cost-effective (depending on the business need). The sensors could send data using machine-to-machine communication to a central node device, which is on the edge and has the capability to run ML algorithms or do predictive analytics in conjunction with the backend cloud.

There is cost involved but it is a tradeoff in terms of the outcomes and the ROI that you expect. If there is a proper balance of security, cost, and

form factor, then the edge could be very secure.

According to Venkateswar Melachervu, there is no drastic change in hardware on the edge from what is implemented on the cloud; they are pretty much the same. Edge devices, particularly mobile devices, have to run one or two iterations. This generally causes the battery to drain very fast, which needs to be recharged frequently. This is a constraint in terms of battery life as the device has to run recursively and there is a lot of computational load.

Purpose-built AI on edge

The stack that defines edge could be divided into several parts: Hardware, AI-capable microcontrollers, AI cores (CNNs, BNNs, accelerators, and other neural network binary codes), whether it is open-source or from Google. And then there are software tools that could cross-compile high-level C++ or Java code onto these computing cores, and some reference designs and architectures and custom